3D-Printed Functional Prosthetic Hand

Engineering Design Case Study & Mechanical Systems Architecture

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EXECUTIVE SUMMARY

This engineering case study details the end-to-end design, rapid prototyping, and electromechanical integration of an accessible, low-cost functional prosthetic hand prototype. Developed within the *Fundamentals of Engineering Design* framework at the New Jersey Institute of Technology, the primary design objective was to replicate essential human grasping capabilities while keeping material costs below **$30** and structural weight under **1.5 lbs**.

The resulting mechanical architecture utilizes modular 3D-printed PLA digits driven by a high-torque micro-servo actuator and routed tendon cables. The final prototype successfully achieved a rapid **1.0-second closing speed** and sustained a functional load capacity of up to **370 grams**.



Figure 1: Complete functional prototype assembly with routed tendons and forearm enclosure.

Core Impact & Application

To address global prosthetic accessibility—where upper-limb prosthetics remain cost-prohibitive for millions in developing regions—this project demonstrates a modular, highly scalable design that can be fabricated using desktop 3D printers and off-the-shelf electronics.

TECHNICAL SPECIFICATIONS & CONSTRAINTS

METRIC / ENGINEERING PARAMETER	TARGET SPECIFICATION	ACHIEVED PERFORMANCE	STATUS
Total Device Weight	< 1.50 lbs (680 g)	1.50 lbs total mass	Target Met
Manufacturing Cost	< $30.00 USD	< $30.00 (PLA + Servo + Nano)	Target Met
Grip Actuation Speed	Rapid closing response	1.0 Second full closure	Target Met
Payload Capacity	Everyday household items	Up to 370 g payload strength	Target Met
Structural Material	PLA Filament (3D Printed)	High-density PLA Shell	Target Met
Actuation Method	Servo-driven tendon cables	High-Torque Servo + Fishing Line	Target Met

SYSTEM DESIGN & MECHANICAL ARCHITECTURE

The prosthetic device is divided into three interconnected structural subsystems, optimized in SolidWorks for additive manufacturing and stress distribution:

1. Modular Finger Linkages

Each finger consists of three interlocking segments (tip, mid, and lower) held together by custom 3D-printed pin joints. Embedded internal channels guide nylon tendons for active flexion, while elastic back-bands supply the return force for opening.

2. Palm Base Anchor

Serves as the central mounting chassis for all five digits. Features angled guide orifices to route tendons smoothly into the forearm without mechanical binding.

3. Forearm Electronics Housing

A hollow ergonomic enclosure (7.00" internal length) housing the main servo motor, breadboard, power battery pack, and Arduino Nano microcontroller.

4. Actuation Circuitry

Powered by a high-torque servo motor pulling five routed tendon lines simultaneously to execute a unified power grasp.

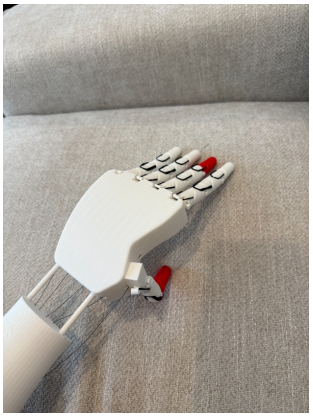


Figure 2: Fully assembled 3D-printed palm and finger assembly showcasing anatomical joint alignment.



Figure 3: Fully assembled 3D-printed palm and finger assembly showcasing anatomical joint alignment. (front)

ENGINEERING ITERATIONS & PROBLEM SOLVING

- **Joint Pin Tolerance & Friction:** Initial metal pins caused excessive binding or excessive slack. Custom compressible 3D-printed snap pins were engineered, maintaining tight tolerances while allowing smooth rotation.
- **Thumb Opposition Angle:** Early CAD prints placed the thumb parallel to the fingers, preventing effective grasping. The thumb joint was re-engineered at an offset angle to allow anatomical opposition against the index and middle fingers.
- **Tendon Route Friction:** Internal palm channels were chamfered and smoothed to minimize cable friction, enabling an efficient 1.0s power grasp response.

FUTURE ROADMAP

While the current prototype meets all FED 101 requirements through button/switch triggers, future engineering enhancements include:

- **Myoelectric Control (EMG):** Integrating surface EMG sensors to detect forearm muscle contractions for intuitive control.
- **Compliant Grip Pads:** Adding TPU or silicone fingertips to reduce slip on smooth curved surfaces.
- **Fiber-Reinforced Printing:** Utilizing PETG or Carbon Fiber PLA for increased structural durability.

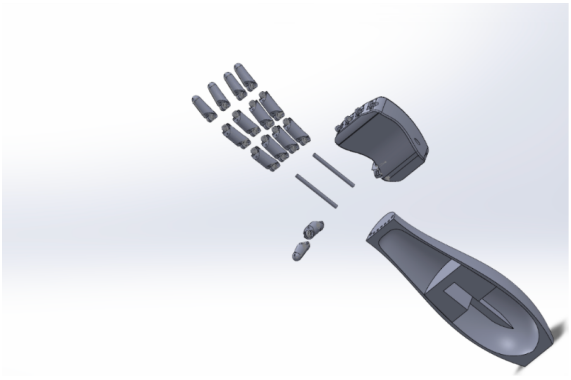


Figure 2: Exploded SolidWorks CAD model detailing components.

VIDEO DEMOSTRATION LINK

https://drive.google.com/file/d/1cMehkLD05mA7CgPNLcuOTKP0tDlBN7R5/view?usp=drivesdk